AI Labeling – EU AI Act Art. 50 for TYPO3

`4viewture/ai-labeling` (extension key `ai_labeling`) adds transparency fields for AI-generated or AI-modified content to **file metadata** (`sys_file_metadata`), **content elements** (`tt_content`) and **all other eligible record tables** (pages, news, …), following the guide „KI-generierte Inhalte kennzeichnen“ (Mittelstand-Digital Zentrum Berlin). Transparency obligations apply from 2 August 2026.

Supports **TYPO3 12.4 LTS, 13.4 LTS and 14**, PHP 8.1+.

Editor guide (German): Documentation/Leitfaden.md

Example file in file_metadata

Datei-Metadaten "Barrierefreiheitserklaerung_im_digitalen_Design.pn..." auf Wurzelebene bearbeiten

Allgemein | Metadaten | Kameradaten | Zugriff | Kategorien | Suche | KI | Export

KI-Beteiligung

Einsatz von KI `[tx_ailabeling_involvement]`

Technische Anpassungen wie Farbkorrektur, Hintergrundgestaltung, Skalierung, Schärfung, Komprimierung oder Rechtschreibprüfung gelten als „KI-unterstützt" und sind nicht kennzeichnungspflichtig.

Mit KI erzeugt (AI GENERATED) [3]

Automatisch erkannt

Die Metadaten der Datei weisen auf KI-Erzeugung oder -Bearbeitung hin. Die Datei wurde automatisch gekennzeichnet – bitte die Fragen unten prüfen und ggf. anpassen.

```
IPTC DigitalSourceType: trainedAlgorithmicMedia
C2PA Content Credentials
```

▸ **Leitfaden anzeigen: KI-Inhalte kennzeichnen (Art. 50 KI-Verordnung)**

KI-Werkzeug / Modell `[tx_ailabeling_tool]`

z. B. Midjourney v7, ChatGPT, Adobe Firefly – zur Nachvollziehbarkeit.

Deepfake-Prüfung (Bild, Audio, Video)

Bezug zu real existierenden oder plausiblen Personen, Orten oder Ereignissen `[tx_ailabeling_real_reference]`

Eine bekannte Person ist nicht nötig. Auch realistisch wirkende, erfundene Menschen oder Orte zählen.

[0]

Kann für echt gehalten werden (Täuschungspotenzial) `[tx_ailabeling_deceptive]`

Fantasie-, Comic- oder erkennbar stilisierte Motive, Icons, Infografiken und abstrakte Grafiken haben in der Regel kein Täuschungspotenzial.

[0]

KI-Person tritt als echte Kundin, echter Kunde oder Zeuge auf (Testimonial) `[tx_ailabeling_synthetic_person]`

Gilt als Deepfake und muss gekennzeichnet werden. Zusätzlich besteht ein wettbewerbsrechtliches Risiko.

[0]

KI verändert das Produkt selbst oder seine Nutzung `[tx_ailabeling_product_altered]`

z. B. Farbe, Material, Größe, Zubehör oder Nutzungssituation. Irreführende Darstellungen können auch mit Kennzeichnung wettbewerbsrechtlich problematisch sein.

[0]

Kennzeichnung

Kennzeichnung `[tx_ailabeling_mode]`

„Automatisch" folgt der Einschätzung oben (kennzeichnet, wenn erforderlich oder empfohlen). Im Zweifel kennzeichnen – freiwillig ist das jederzeit möglich.

Immer kennzeichnen [1]

Kennzeichnungstext `[tx_ailabeling_label_text]`

Wird im Frontend mit dem Inhalt angezeigt. Leer lassen für den Standardtext.

Standardtext, z. B. „Dieser Text wurde mit Hilfe von KI erzeugt."

Interne Notizen `[tx_ailabeling_notes]`

Automatische Erkennung

Automatisch aus Datei-Metadaten erkannt `[tx_ailabeling_detected]`

[1]

Erkennungsmerkmale `[tx_ailabeling_detection]`

IPTC DigitalSourceType: trainedAlgorithmicMedia
C2PA Content Credentials

Datei-Metadaten [5976] `[sys_file_metadata]`

Features

- Own **„KI“ tab** on every type of every eligible table – columns, palettes and tab are added in `AfterTcaCompilationEvent` (tables/types of extensions loaded later are covered), database columns via `AlterTableDefinitionStatementsEvent` (`sys_file_metadata` / `tt_content` also in `ext_tables.sql`).
- **Eligible tables** (`tables = *`): all tables with types, except `be_*` , `sys_*` (but `sys_file_metadata` is always included), `fe_users` , `fe_groups` , cache/index/scheduler/impexp tables, `adminOnly` , `readOnly` , `is_static` and root-level (`rootLevel = 1`) tables. Restrict with `tables = tt_content,pages,…` or `excludeTables` . Flush caches and run the database compare after changing it.
- **Record icon overlay** for all labeled tables (page module, list module, form header, element browser …): `ModifyRecordOverlayIconIdentifierEvent` (13+) / `overrideIconOverlay` hook (12). Core status overlays (hidden, start/endtime, access, deleted) keep priority. The list module only selects visible columns, so `ModifyDatabaseQueryForRecordListingEvent` adds the two needed fields. Rows without the fields (e.g. page tree) get no overlay – no extra queries.
- Decision questions from the guide (deepfake criteria, testimonial, product altered, public-interest text, human review) with field descriptions.
- **Live assessment** (`<ai-labeling-guide>` web component) on top of the tab: required / recommended / optional / none, reasons, resulting label – plus the collapsible guide.
- Label mode: automatic / always / never (warning flash message if a required label is switched off).
- Computed, persisted fields for the frontend: `tx_ailabeling_status` , `tx_ailabeling_active` , `tx_ailabeling_reasons` (calculated on save by a DataHandler hook).
- **Automatic detection on upload/replace** (`AfterFileAddedEvent` , `AfterFileReplacedEvent`):
  - IPTC Digital Source Type in XMP/C2PA (`trainedAlgorithmicMedia` , `compositeWithTrainedAlgorithmicMedia` , `compositeSynthetic` , `algorithmicallyEnhanced`)
  - C2PA / Content Credentials claim generator
  - PNG text chunks of Stable Diffusion WebUI, ComfyUI, InvokeAI, NovelAI
  - known AI tool names in metadata only (XMP, EXIF, PNG text, ID3, PDF info, MP4 `©too`)
- **File list indicators**
  - Icon overlay via `ModifyIconForResourcePropertiesEvent` (12.4–14): red AI star = labelled, outlined star = AI generated/modified but not labelled. Shown wherever the core renders file icons (file list, element browser, file references, live search). Core overlays (e.g. missing file) win.
  - Columns *AI labelling status* / *AI label active* selectable in the column selector of the file list (metadata fields, readOnly – not part of the edit form).
  - TYPO3 14+: text badge in the name column and `data-ai-labeling*` row attributes via `AfterFileListRowPreparedEvent` .
- CLI / scheduler command for existing files: `vendor/bin/typo3 ailabeling:scan [--dry-run] [--storage=1] [--limit=500] [--override-manual]`

Installation

```
composer require 4viewture/ai-labeling
vendor/bin/typo3 extension:setup
```

Fields are `exclude => true` – grant them to editor groups (*Allowed excludefields*).

Extension configuration

Setting	Default	Meaning
`tables`	`*`	Tables with AI fields (`*` or comma list)
`excludeTables`	–	Tables to skip
`autoDetect`	1	Analyse files on upload/replace
`autoLabel`	1	Set label mode of detected files to "always label"
`scanLimitMb`	32	Bytes read per file (head + tail half each for larger files)
`additionalSignatures`	–	Comma-separated extra tool names

Fields

Field	Type	Notes
`tx_ailabeling_involvement`	int	0 none, 1 assisted, 2 modified (AI MODIFIED), 3 generated (AI GENERATED)
`tx_ailabeling_tool`	string	tool / model
`tx_ailabeling_real_reference` , `_deceptive` , `_synthetic_person` , `_product_altered`	bool	deepfake / product questions
`tx_ailabeling_public_interest` , `_human_review` , `_reviewer`	bool/string	text questions
`tx_ailabeling_mode`	int	0 auto, 1 always, 2 never
`tx_ailabeling_label_text`	string	empty = default text (`LabelTextResolver`)
`tx_ailabeling_detected` , `_detection`	bool/text	automatic detection + evidence
`tx_ailabeling_status`	int	computed: 0 none, 1 optional, 2 recommended, 3 required
`tx_ailabeling_active`	bool	computed: **render a label in the frontend if 1**
`tx_ailabeling_reasons`	csv	computed reason keys

Architecture

```
Classes/
  Configuration/TcaFields.php        shared TCA (columns, palettes, tab, SQL)
  Configuration/LabelingTables.php   which tables get the fields
  Detection/AiContentDetector.php    TYPO3-independent metadata analysis
  Service/LabelEvaluator.php         decision logic of the guide (mirrored in label-guide.js)
  Service/AutoLabeler.php            detector -> FAL metadata
  Service/LabelTextResolver.php      label text / defaults (for the upcoming frontend part)
  EventListener/                     TCA/SQL for all tables, detection, icon overlays, list module se
  Service/RecordOverlay.php          overlay decision for record icons
  Hooks/DataHandlerHook.php          persist computed status on save
  Form/Element/GuideInformation.php  FormEngine fieldWizard: guide + live assessment
  Command/ScanCommand.php            ailabeling:scan
```

Compatibility notes: no PHP attributes for listeners/commands (v12), no Fluid `StandaloneView` (removed in v14), FormEngine node without own constructor (v12 constructor / v13+ `setData()`), file types as integers instead of `AbstractFile::FILETYPE_*`.

Limitations

- Detection is heuristic. Metadata is often stripped by editing tools or social networks; invisible watermarks (e.g. SynthID) cannot be read. A tool name inside a human-written caption can cause a false positive – the editor reviews the result in the AI tab.
- Detection writes the default-language metadata record only.
- Tiles view with thumbnails: the thumbnail replaces the icon, so the overlay is only visible for files without preview; the core offers no event for tiles (12–14).
- Record overlay only when the core shows no status overlay (hidden etc. win) and only where the rendered row contains the AI fields (not in the page tree).
- Frontend output is not part of this version.

Tests

```
vendor/bin/phpunit Tests/Unit
```

The unit tests cover the decision logic and the detector (synthetic JPEG/XMP, PNG chunks, C2PA/CBOR, large files).